

Computer Science I

Chapter 4 - Making Decisions

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Slide contents are mainly based on the following documents:

- Tony Gaddis, Starting out with C++, Six Edition, Pearson, 2009
- Tony Gaddis, Power Points Slides:
<https://www.ums1.edu/~antognolij/gaddis.html>



Logical Operators

- Used to create relational expressions from other relational expressions

Operator	Meaning	Description
&&	AND	New relational expression is true if both expressions are true
	OR	New relational expression is true if either expression is true
!	NOT	Reverses the value of an expression – true expression becomes false, and false becomes true



Logical Operators

- `int x = 12, y = 5, z = -4;`

<code>(x > y) && (y > z)</code>	<code>true</code>
<code>(x > y) && (z > y)</code>	<code>false</code>
<code>(x <= z) (y == z)</code>	<code>false</code>
<code>(x <= z) (y != z)</code>	<code>true</code>
<code>!(x >= z)</code>	<code>false</code>



Checking Numeric Ranges with Logical Operators

- Used to test to see if a value falls **inside** a range:

```
if (grade >= 0 && grade <= 100)  
    cout << "Valid grade";
```
- Can also test to see if value falls **outside** of range:

```
if (grade <= 0 || grade >= 100)  
    cout << "Invalid grade";
```
- Cannot use mathematical notation:

```
if (0 <= grade <= 100) //doesn't work!
```



Validating User Input

- Input validation: inspecting input data to determine whether it is acceptable
- Bad output will be produced from bad input
- Can perform various tests:
 - Range
 - Reasonableness
 - Divide by zero



Validating User Input

```
16     int testScore; // To hold a numeric test score
17
18     // Get the numeric test score.
19     cout << "Enter your numeric test score and I will\n"
20          << "tell you the letter grade you earned: ";
21     cin >> testScore;
22
23     // Validate the input and determine the grade.
24     if (testScore >= MIN_SCORE && testScore <= MAX_SCORE)
25     {
26         // Determine the letter grade.
27         if (testScore >= A_SCORE)
28             cout << "Your grade is A.\n";
29         else if (testScore >= B_SCORE)
30             cout << "Your grade is B.\n";
31         else if (testScore >= C_SCORE)
32             cout << "Your grade is C.\n";
33         else if (testScore >= D_SCORE)
34             cout << "Your grade is D.\n";
35         else
36             cout << "Your grade is F.\n";
37     }
38     else
39     {
40         // An invalid score was entered.
```



Comparing Characters

- Characters are compared using their ASCII values
- 'A' < 'B': The ASCII value of 'A' (65) is less than the ASCII value of 'B' (66)
- '1' < '2': The ASCII value of '1' (49) is less than the ASCII value of '2' (50)
- Lowercase letters have higher ASCII codes than uppercase letters, so 'a' > 'Z'



ASCII table

32: SPC	33: !	34: "	35: #	36: \$
37: %	38: &	39: '	40: (	41:)
42: *	43: +	44: ,	45: -	46: .
47: /	48: 0	49: 1	50: 2	51: 3
52: 4	53: 5	54: 6	55: 7	56: 8
57: 9	58: :	59: ;	60: <	61: =
62: >	63: ?	64: @	65: A	66: B
67: C	68: D	69: E	70: F	71: G
72: H	73: I	74: J	75: K	76: L
77: M	78: N	79: O	80: P	81: Q
82: R	83: S	84: T	85: U	86: V
87: W	88: X	89: Y	90: Z	91: [
92: \	93:]	94: ^	95: _	96: `
97: a	98: b	99: c	100: d	101: e
102: f	103: g	104: h	105: i	106: j
107: k	108: l	109: m	110: n	111: o
112: p	113: q	114: r	115: s	116: t
117: u	118: v	119: w	120: x	121: y
122: z	123: {	124:	125: }	126: ~
127: DEL				



Comparing Characters

```
10 // Get a character from the user.
11 cout << "Enter a digit or a letter: ";
12 ch = cin.get();
13
14 // Determine what the user entered.
15 if (ch >= '0' && ch <= '9')
16     cout << "You entered a digit.\n";
17 else if (ch >= 'A' && ch <= 'Z')
18     cout << "You entered an uppercase letter.\n";
19 else if (ch >= 'a' && ch <= 'z')
20     cout << "You entered a lowercase letter.\n";
21 else
22     cout << "That is not a digit or a letter.\n";
```



Comparing string Objects

- Like characters, strings are compared using their ASCII values

```
string name1 = "Mary";  
string name2 = "Mark";
```

```
name1 > name2 // true  
name1 <= name2 // false  
name1 != name2 // true
```

```
name1 < "Mary Jane" // true
```



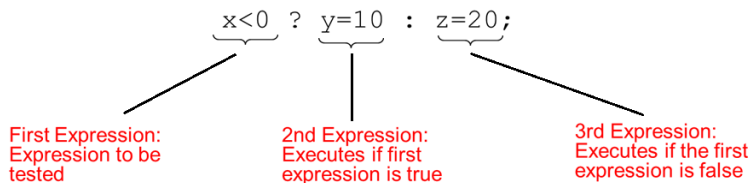
Comparing string Objects

```
26 // Determine and display the correct price
27 if (partNum == "S-29A")
28     cout << "The price is $" << PRICE_A << endl;
29 else if (partNum == "S-29B")
30     cout << "The price is $" << PRICE_B << endl;
31 else
32     cout << partNum << " is not a valid part number.\n";
```



The Conditional Operator

- Can use to create short if/else statements
- Format: `expr ? expr : expr;`



The Conditional Operator

- The value of a conditional expression is
 - The value of the second expression if the first expression is true
 - The value of the third expression if the first expression is false
- Parentheses () may be needed in an expression due to precedence of conditional operator



The Conditional Operator

```
1 // This program calculates a consultant's charges at $50
2 // per hour, for a minimum of 5 hours. The ?: operator
3 // adjusts hours to 5 if less than 5 hours were worked.
4 #include <iostream>
5 #include <iomanip>
6 using namespace std;
7
8 int main()
9 {
10     const double PAY_RATE = 50.0; // Hourly pay rate
11     const int MIN_HOURS = 5;      // Minimum billable hours
12     double hours,                 // Hours worked
13           charges;                // Total charges
14
15     // Get the hours worked.
16     cout << "How many hours were worked? ";
17     cin >> hours;
18
19     // Determine the hours to charge for.
20     hours = hours < MIN_HOURS ? MIN_HOURS : hours;
21
22     // Calculate and display the charges.
23     charges = PAY_RATE * hours;
24     cout << fixed << showpoint << setprecision(2)
25          << "The charges are $" << charges << endl;
26     return 0;
27 }
```



The switch Statement

- Used to select among statements from several alternatives
- In some cases, can be used instead of if/else if statements

```
switch (expression) //integer
{
    case exp1:  statement1;
    case exp2:  statement2;
    ...
    case expn:  statementn;
    default:   statementn+1;
}
```



switch Statement Requirements

- expression must be an integer variable or an expression that evaluates to an integer value
- exp1 through expn must be constant integer expressions or literals, and must be **unique** in the switch statement
 - case (x < 20): // does not work
 - case (n == 20): // does not work
- default is optional but recommended



switch Statement-How it Works

1. expression is evaluated
2. The value of expression is compared against exp1 through expn.
3. If expression matches value exp_i, the program branches to the statement following exp_i and continues to the end of the switch
4. If no matching value is found, the program branches to the statement after default



The switch Statement

Program 4-23

```
1 // The switch statement in this program tells the user something
2 // he or she already knows: the data just entered!
3 #include <iostream>
4 using namespace std;
5
6 int main()
7 {
8     char choice;
9
10    cout << "Enter A, B, or C: ";
11    cin >> choice;
12    switch (choice)
13    {
14        case 'A': cout << "You entered A.\n";
15                  break;
16        case 'B': cout << "You entered B.\n";
17                  break;
18        case 'C': cout << "You entered C.\n";
19                  break;
20        default:  cout << "You did not enter A, B, or C!\n";
21    }
22    return 0;
23 }
```

Program Output with Example Input Shown in Bold

Enter A, B, or C: **B [Enter]**
You entered B.

Program Output with Example Input Shown in Bold



break Statement

- Used to exit a `switch` statement
- If it is left out, the program "falls through" the remaining statements in the `switch` statement



The switch Statement

Program 4-25

```
1 // This program is carefully constructed to use the "fall through"
2 // feature of the switch statement.
3 #include <iostream>
4 using namespace std;
5
6 int main()
7 {
8     int modelNum; // Model number
9
10    // Get a model number from the user.
11    cout << "Our TVs come in three models:\n";
12    cout << "The 100, 200, and 300. Which do you want? ";
13    cin >> modelNum;
14
15    // Display the model's features.
16    cout << "That model has the following features:\n";
17    switch (modelNum)
18    {
19        case 300: cout << "\tPicture-in-a-picture.\n";
20        case 200: cout << "\tStereo sound.\n";
21        case 100: cout << "\tRemote control.\n";
22                break;
23        default: cout << "You can only choose the 100,";
24                cout << "200, or 300.\n";
25    }
26    return 0;
27 }
```



The switch Statement

Program Output with Example Input Shown in Bold

Our TVs come in three models:

The 100, 200, and 300. Which do you want? **100 [Enter]**

That model has the following features:

Remote control.

Program Output with Example Input Shown in Bold

Our TVs come in three models:

The 100, 200, and 300. Which do you want? **200 [Enter]**

That model has the following features:

Stereo sound.

Remote control.

Program Output with Example Input Shown in Bold

Our TVs come in three models:

The 100, 200, and 300. Which do you want? **300 [Enter]**

That model has the following features:

Picture-in-a-picture.

Stereo sound.

Remote control.

Program Output with Example Input Shown in Bold

Our TVs come in three models:

The 100, 200, and 300. Which do you want? **500 [Enter]**

That model has the following features:

You can only choose the 100, 200, or 300.



Using switch in Menu Systems

- `switch` statement is a natural choice for menu-driven program:
 - display the menu
 - then, get the user's menu selection
 - use user input as `expression` in `switch` statement
 - use menu choices as `expr` in case statements



More About Blocks and Scope

- Scope of a variable is the block in which it is defined, from the point of definition to the end of the block
- Usually defined at beginning of function
- May be defined close to first use



More About Blocks and Scope

```
16     if (income >= MIN_INCOME)
17     {
18         // Get the number of years at the current job.
19         cout << "How many years have you worked at "
20             << "your current job? ";
21         int years;        // Variable definition
22         cin >> years;
23
24         if (years > MIN_YEARS)
25             cout << "You qualify.\n";
26         else
27         {
28             cout << "You must have been employed for\n"
29                 << "more than " << MIN_YEARS
30                 << " years to qualify.\n";
31         }
32     }
```



Variables with the Same Name

- Variables defined inside $\{ \}$ have local or block scope
- When inside a block within another block, can define variables with the same name as in the outer block.
 - When in inner block, outer definition is not available
 - Not a good idea



Variables with the Same Name

Program 4-30

```
1 // This program uses two variables with the name number.
2 #include <iostream>
3 using namespace std;
4
5 int main()
6 {
7     // Define a variable named number.
8     int number;
9
10    cout << "Enter a number greater than 0: ";
11    cin >> number;
12    if (number > 0)
13    {
14        int number; // Another variable named number.
15        cout << "Now enter another number: ";
16        cin >> number;
17        cout << "The second number you entered was "
18             << number << endl;
19    }
20    cout << "Your first number was " << number << endl;
21    return 0;
22 }
```

Program Output with Example Input Shown in Bold

```
Enter a number greater than 0: 2 [Enter]
Now enter another number: 7 [Enter]
```



- Programming challenges: 13 - 18
- A program that asks users to enter two numbers (a and b) and an operator (+, -, *, /), then prints the result of applying that operator to the two numbers.
- A program that asks users to enter 4 numbers (a, b, c, d) and uses conditional operators to find the maximum of them.
- Write a program that asks the user to enter a two-digit integer and outputs the number in Vietnamese words.
- A program that asks users to enter an integer and outputs the hundreds digit of that number. If there is no hundreds digit, output 0.
- A program that asks the user to enter any valid date (day, month) in 2025 and determines what day of the week it is.

